

Full comparison-contract admission for the labelled PAH-v2 model

TECT verification-first repository
claim C6-SPACETIME-SIGNATURE

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1. Purpose and scope

The operator has adopted the complete Morton/CRT comparison bundle for research use. This note checks that exact contract, not a replacement model. The immutable accepted bundle has SHA-256

1d6fa47f5351d74f05bed3aba5b1e229a827644d4c6ef5d7dd73c117adee95b6

Its dated approval and operative reference are [strategy/owner-decisions/PAH-v2-morton-crt-approval-20260912.json](#) and [strategy/pa-hyp/PAH-v2-morton-crt-contract-v1.json](#). Earlier unapproved labels are historical issuance metadata, not rewritten. Approval activates definitions, not their mathematical correctness.

All statements concern the full finite counting spaces of fixed PAH-001-v2-r2, with R-570 retained as a finite-source reference only. The functional, labelled rates, mobility exponent, state normalization, projection and external Markov time are unchanged. No generator comparison experiment, limiting generator domain or physical claim is made. The admitted algebra is comparison-defined: terminal occupation does not mean literal preservation of the same site's value.

2. Setting and precise finite-definition statement

Fix integers $1 \leq q_0 \leq m_0$, $R_0 > 0$ and the parent's real parameters. For $\rho = (r, h, N)$, the graph has side $W = 2^{h+N+1}$, vertices in $\{0, \dots, W-1\}^2$, every horizontal and vertical nearest-neighbor edge, and every unit square face. Stored positive edge orientations determine signs. The anchors are $O = (0, 0)$ and $C = (2^h, 0)$; no fields are clamped. Degree is at most four. These are combinatorial, not physical, coordinates.

Let K_r be the product of the first $r+1$ primes and set

$$M_{s,r} = 2^r, \quad M_{\psi,r} = m_0 4^r, \quad R_{\max,r} = R_0 2^r, \quad Q_r = q_0 2^r.$$

All cutoffs are unbounded and $Q_r \leq M_{\psi,r}$. Every integer tuple (j, ℓ, n, u) in the parent's ranges with total occupation Q_r remains a state, including hidden labels at $\ell = 0$ and $\epsilon = 1$. Set $A_\rho = \mathbb{C}^{\Omega_\rho}$ and let A_ρ^{inv} be its full gauge/anchor-cell invariant subalgebra.

Theorem 1. *On this exact family, the approved maps $p_{\sigma,\rho} : \Omega_\sigma \rightarrow \Omega_\rho$ are total, surjective and compose for every $\rho \leq \sigma$ in product order. Their pullbacks are unital $*$ -algebra embeddings preserving the full invariant subalgebras and inherited supremum norm, with the declared finite support bound. Directed-root assignments retain unpaired channels and are inverse-coherent where paired.*

This is not $L_\sigma I = I L_\rho$, an infinite-dynamics domain theorem, or state convergence. The original local-cutoff, lattice, exhaustion, then inherited later-limit order is retained; finite composition does not interchange limits.

3. Independent arbitrary-pair formula and proof

For $\rho = (r, h, N) \leq \sigma = (s, k, M)$ put

$$D = 2^{s-r}, \quad H = 2^{k-h}, \quad b = H^2, \quad P = K_s/K_r, \quad A = P^{-1} \pmod{K_r}.$$

The primes in P do not divide K_r , so the inverse exists. In Morton order the representative of coarse z is bz and its descendants are the contiguous block $bz, \dots, b(z+1) - 1$. A retained refined box is a prefix of the fine box. Put $v = |V_\rho|$, $t = v - 1$, and use exclusive prefixes $S_a = \sum_{i < a} \ell_i$ of the fine tuple. Then

$$\begin{aligned} \ell'_z &= \lfloor S_{b(z+1)}/D \rfloor - \lfloor S_{bz}/D \rfloor \quad (z < t), & \ell'_t &= Q_r - \lfloor S_{bt}/D \rfloor, \\ j'_z &= \lfloor j_{bz}/D \rfloor, & n'_z &= An_{bz} \pmod{K_r}, & u'_e &= A \sum_{a \in \gamma_H(e)} u_a \pmod{K_r}. \end{aligned}$$

Here $\gamma_H(e)$ is the signed axial path of length H between representatives. These are expressions for the approved compositions, not new definitions.

Totality and surjectivity. Prefix monotonicity gives nonnegative differences. Including the terminal complement they sum to Q_r , so each is at most $Q_r \leq M_{\psi,r}$. Since $S_{bt} \leq Q_s = DQ_r$, the terminal value is valid. All other ranges hold. For any coarse tuple put $Dj_z, D\ell_z, Pn_z$ at bz , zero at other fine vertices, and Pu_e on the first edge of $\gamma_H(e)$, zero on its other edges. Distinct coarse paths share no fine edge. This is valid because $DM_{s,r} = M_{s,s}$, $DM_{\psi,r} \leq M_{\psi,s}$, and $DQ_r = Q_s$. Substitution gives $pJ = \text{id}$ on every coarse tuple. This injection does not condition the actual fine ensemble or replace its Gibbs normalization.

Composition for every comparable triple. Every nonterminal output prefix is the corresponding fine prefix divided by D and floored; the complete output prefix is Q_r . Grouping contiguous blocks telescopes differences. The identity $\lfloor \lfloor a/d \rfloor / e \rfloor = \lfloor a/(de) \rfloor$ handles successive cutoffs and cutoff/block squares. Terminal complements use the same total and the same nonterminal prefix on either route, handling cutoff/exhaustion and lattice/exhaustion squares. Representatives multiply their H factors; paths concatenate; modular multipliers become the inverse of the product prime ratio modulo the final K_r . They commute with signed sums. Aperture floors commute with sampling. This proves the arbitrary-pair formula, identity and composition for all comparable triples, not just the executed sample cube.

Boundary transport is not surjective on faces. Each coarse face maps to four fine faces at an adjacent lattice step, with internal signed edges cancelling. The fine graph also contains an outer right/top fringe:

$$(2W - 1)^2 - 4(W - 1)^2 = 4W - 3.$$

All these additional fine faces remain in the source functional and carrier. No requirement that the cellular transport be onto faces is imposed. Literal exhaustion keeps all old faces. The approved contract does not identify a fixed physical-volume target; it is not pure subdivision of an identical physical boundary.

4. Symmetry, algebra, support and roots

Gauge transformation adds g_v to n_v and $g_w - g_v$ to $u_{(v,w)}$. The coarse label is Ag_{bz} modulo K_r . Gauge increments telescope along each path, giving $p(gx) = g'p(x)$ for every pair. This modular projection is an additive group map, not generally a unital ring map or small-angle rounding.

The full anchored graph group on this family is trivial. Four degree-two corners are intrinsic. Distances from O identify the opposite corner; the two adjacent corners have distinct distances $W - 1 - 2^h$ and $W - 1 + 2^h$ from C . The pair $(x + y, W - 1 - x + y)$ then uniquely determines every vertex. Thus the actual full cell group is used, not a conveniently chosen subgroup. No all-carrier symmetry theorem follows.

For $If = f \circ p$, linearity, unit, products and conjugation follow by evaluation. Gauge equivariance preserves all invariant observables. Surjectivity gives

$$\sup_{x \in \Omega_\sigma} |f(px)| = \sup_{y \in \Omega_\rho} |f(y)|.$$

The algebraic direct limit of the full invariant algebras has this representative-independent norm. Formulas equal on Ω_ρ cannot be distinguished by pullback. No Gibbs norm or reduced observable set is used.

If f is represented at (r, h, N) and the target is (s, k, M) , all output coordinates depend only on vertices and internal edges of $G_{k,N}$. The terminal coordinate is total Q_s minus a retained prefix. This whole refined source box is a computable support for all $M \geq N$. Use $N_{\text{support}}(f) = N$ at fixed target k . This need not be minimal support, a uniform metric-radius bound, or a generator stabilization index.

The lesser endpoint/root key chooses one orientation of each fine incidence and its inverse. Select the first original valid coarse label joining projected endpoints, or UNPAIRED. Reversal chooses the same key and reverses that coarse label. All fine labels, duplicate $K = 2$ inverse labels and unmatched coarse labels remain. This is bookkeeping, not a gauge-equivariant root-Hilbert intertwiner or rate equality. The measure is still $\pi(x)$ per directed root, with $\sqrt{c}/2$ in B .

5. Numerical verification and Lean scope

The admission preregistration is [strategy/pa-hyp/PAH-v2-morton-crt-admission-prereg.json](#). Primary uses the adopted adjacent-map implementation. Independent code imports neither it nor the source root enumerator; coordinate dictionaries and every D -th charge token implement the direct formula. Shared inputs and same-task authorship are explicit: this is internal independence, not an external referee.

The declared fixtures give 24 state cases, 144 axis-order comparisons, 48 adjacent-word interleaving checks and 265 retained valid root incidences. Independent checks include 24 intermediate-index factorizations, gauge identities and right inverses, plus 16 exterior-support changes. These are bounded integer checks, not full fine-state exhaustion or an infinite-family proof from tables. The earlier smallest-coarse enumeration is replayed as development evidence only.

Lean 4.32.1 and locked Mathlib check 17 parameterized declarations in [verification/lean/Tect/PahV2Comparison.lean](#): nested floors, scaled right inverses, prefix monotonicity/telescopes, terminal complements, signed additive phase/path algebra, corner-distance injectivity, and generic pullback composition, products, conjugation, injectivity and full sup-bound equivalence. The concrete grid, state encoding, primordial construction, formula correspondence, root assignment and direct-limit construction are not a complete Lean formalization. They require the written argument above and independent checks.

```
python -X utf8 verification/scripts/pah_v2_comparison_primary.py --check
python -X utf8 verification/scripts/pah_v2_comparison_independent.py --check
python -X utf8 verification/scripts/pah_v2_comparison_hostile.py --check
python -X utf8 verification/scripts/pah_v2_comparison_lean.py --check
```

Lean replay accepts explicit `-lean-cache` and `-elan-home`. All runs are in [claims/C6-SPACETIME-SIGNATURE/runs/2026-09-12-pah-v2-comparison/](#).

6. Devil's-advocate review

1. **Ordinary phase rounding commutes with sums. DISMISSED.** The mutation $K_f = 6, K_c = 2, u_1 = u_2 = 2$ fails; the additive projection is required.
2. **Pointwise floor or literal restriction preserves charge. DISMISSED.** Two unit occupations lose charge under separate halving; external charge disappears under literal restriction.

3. **Terminal occupation is preserved literally. UPHELD as a prohibition.** All charge outside gives empty literal terminal site but comparison value Q . This accepted identification is explicitly different.
4. **No new boundary cells are present. UPHELD as a prohibition.** The extra fringe is counted and retained. No physical-volume equality follows.
5. **Unmatched roots can be dropped. DISMISSED.** All labels, unmatched incidences and coarse channels remain; rates are unchanged.
6. **Finite support, sup isometry or Lean proves dynamics. UPHELD as a prohibition.** None implies generator intertwining, Gibbs projectivity or a limiting invariant core. The future defect keeps both labelled generator sums; sup, Gibbs- L^2 and state discrepancy are separate.
7. **Approval is proof or the audit is external review. UPHELD as a prohibition.** Authority and verification have separate pins; internal independent implementation and kernel checks are accurately scoped.

7. Result footer

Record:	EXP-001737; definition admission only
Result ID:	R-571
Precise statement:	Full finite comparison-definition admission
Scope:	Adopted anchored Morton/CRT grid family
Dependencies:	Fixed PAH-001-v2-r2; R-570 reference only
Evidence grade:	Analytic, exact checks, partial Lean
Reproduction command:	python -X utf8 verification/scripts/pah_v2_comparison_verify.py
Expected output:	PAH-V2-COMPARISON ADMISSION: PASS
Falsification gate:	Invalid image/composition/symmetry/support
Tier before / after:	C6 T1 / T1
No-overclaim statement:	No generator, limit or physical result
Next required action:	Specify one full-label generator-defect target
Classification:	AUXILIARY_SUPPORT
Exact target:	Owner-adopted full comparison contract
Evidence:	Analytic + exact executable + parameterized Lean
Source model:	PAH-001-v2-r2 and R-570 preserved
Active T-054 gate:	Unchanged
Claim tier before/after:	C6 T1 / T1
Future defects:	Preregistered, not evaluated
Next question:	What precise generator-defect target is authorized?
No-overclaim:	No dynamics, infinite/continuum or physical result